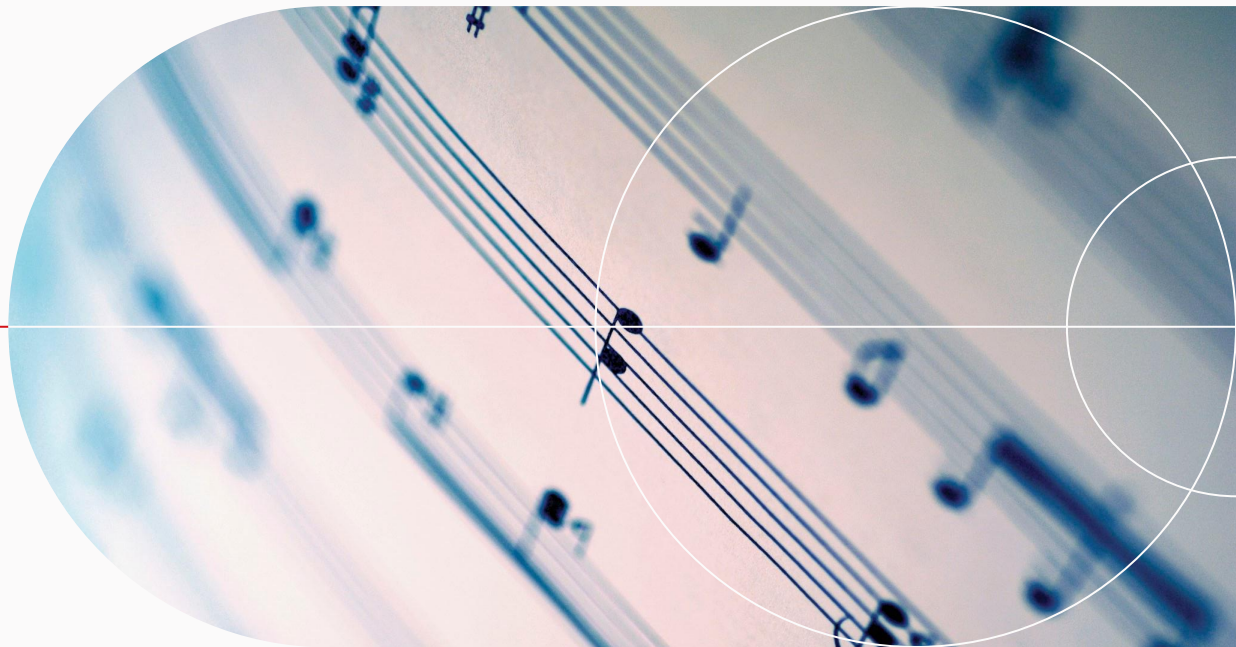


Song Success Prediction Model

By Cameron Hicks





Problem Statement

Music artists, producers, and record labels must make major decisions in the early stages of song production with limited information and high uncertainty. From selecting final versions to be released, to budgets, to targeted regions, these decisions are typically made through intuitive guesses rather than backed by data driven insights.

This project's aim was to develop a predictive machine learning model that can predict a new songs popularity in the pre-release or early release stages.

Data Sources

Spotify Music Analytics Dataset (2015-2025)

- 85,000 Song Records
- 19 Attributes per song including Artist Name, Album Name, Genre, Popularity, Tempo, Danceability, Instrumentalness, Country Name and more
- Provided by Kaggle

Global Data API

- Supplemental Country Data to enrich dataset
- API Data Queried by unique Country Names from Spotify Data
- Features imported include population, languages spoken, currency information, timezones and more

Data Wrangling

Spotify Data

- **Imported** from CSV file
- **Data Types** corrected and columns added for interpretability
 - Dates converted to datetime and track duration added in a reader friendly format
- Inspected for **Duplicate Values**
 - 0 exact duplicates found
 - Several near duplicates found. Inspection showed these were unique re-releases of songs (live, acoustic, etc)
- Inspected for **Missing Values**
 - **Track Name** contained 21 missing values filled with string ***"Unknown"***
 - **Album Name** contained 46 missing values. These records were dropped from the dataset
- Unique **Country Names** were gathered in preparation for QPI Query

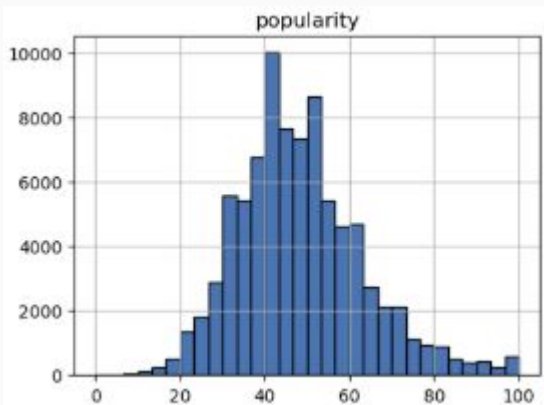
Data Wrangling

Country Data

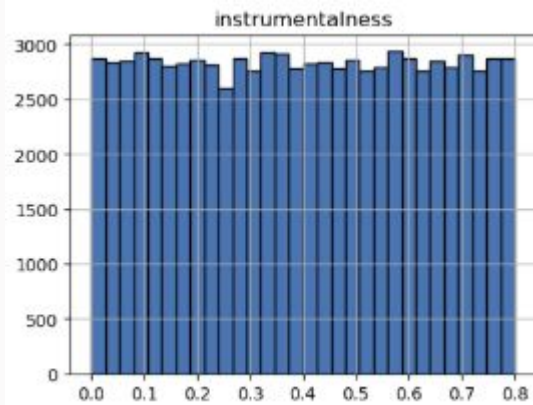
- **Country Data** was queried for the 10 unique Countries in the Spotify Data
- Each Country gained 37 attributes
- Many attributes were comprised of Lists of Lists, or Lists of Dictionaries. Each were inspected, relevant data was identified and the columns were converted to contain relevant data in a way that could be explored, visualized, and train a model.
 - **Timezones** and **Bordering Country** converted to count columns
 - **Currencies** reduced to Currency Code column
 - **Languages** and **regionalBlocs** were One Hot Encoded to boolean columns
- Several attributes were identified as **redundant** or contained **no relevance** to this project. These attributes were dropped.
 - These attributes included Alternative Country Name spellings, Country Codes, Country Flags, Translation Columns (same data translated into a different language)

The Spotify Data and Country Data were merged to create one Combined Data Frame

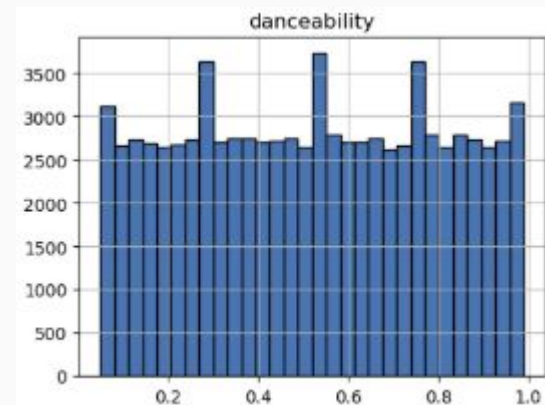
Exploratory Data Analysis



Our Target Feature **Popularity** showed a normal distribution indicating that the majority of the songs have a mid-rank for popularity.



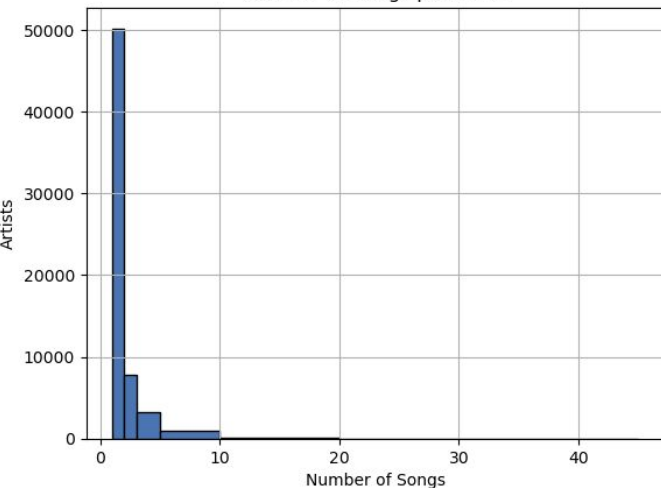
Several other features such as **Instrumentality** and **Danceability** showed a uniform distribution indicating the data set had a wide variety of different features available.



It was surprising that there was not less representation of more lyrical or higher danceability songs as it is a common assumption that music with lyrics and high danceability are more popular.

Exploratory Data Analysis

Number of Songs per Artist

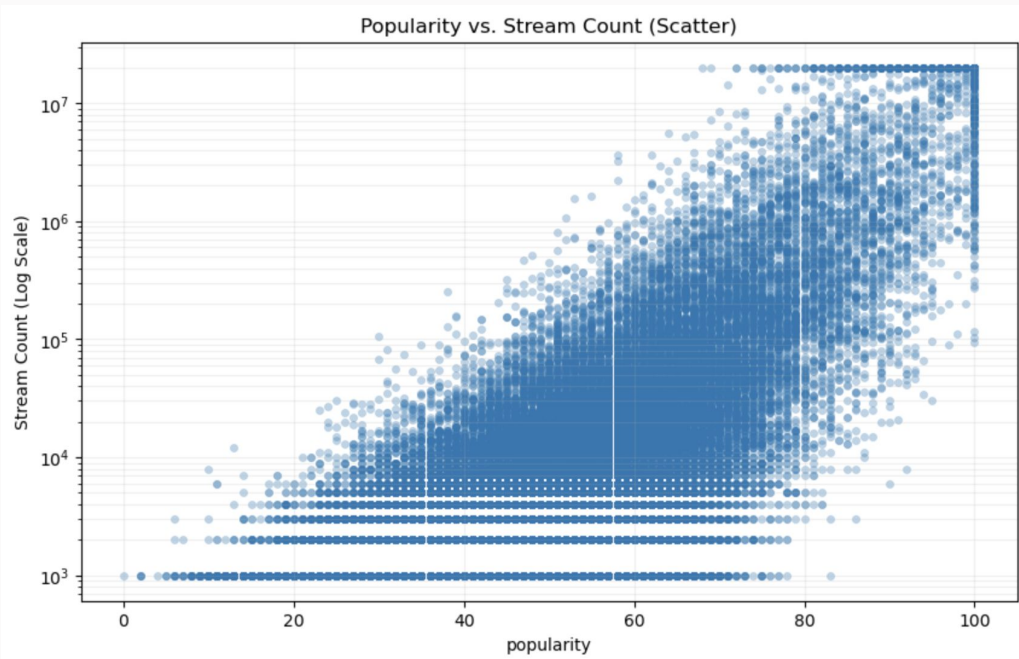


Nearly all **Artists** had less than 10 songs in the dataset, the vast majority only had 1 song represented.

	artist_name	num_songs	num_albums	mean_popularity	median_popularity	total_streams
46066	Nicholas Graham	1	1	100.0	100.0	14458000
58190	Thomas Park	1	1	100.0	100.0	20000000
22635	Harold Price	1	1	100.0	100.0	20000000
38645	Luis Waters	1	1	100.0	100.0	1708000
28522	Jodi Bishop	1	1	100.0	100.0	20000000
19472	Elizabeth Stanley	1	1	100.0	100.0	8383000
25095	Janet Ross	1	1	100.0	100.0	4952000
6646	Bobby Gomez	1	1	100.0	100.0	20000000
21985	Glenn Murphy	1	1	100.0	100.0	20000000
55217	Stanley Baldwin	1	1	100.0	100.0	20000000

Calculating the mean and median popularity scores for artists proved inaccurate due to the large number of artists with only one song. These calculations create leakage of the popularity feature and fail to deliver valuable insight on artist popularity

Exploratory Data Analysis



Stream Count shows a positive relationship to Popularity and was the strongest visible relationship in the EDA process

However, the final model will need to predict popularity prior to a songs release and no stream count data. Training the model on this feature could create a dependency and ultimately inaccurate model

Pre-Processing

- Standard steps for Pre-Processing steps were followed
- Features that contained leakage risk, redundancy, and/or high cardinality were dropped.
 - Included: track_id, track_name, artist_name, album_name, duration copy, latlng, capital, demonym, stream_count, release_date
- One Hot Encoding performed with Pandas get_dummies.
 - Drop First was used to avoid multicollinearity
- Dataset was separated into X and y variables, then split into 80/20 training and testing sets using train_test_split
- Scaling was performed using StandardScaler after splitting the data to prevent leakage. Fit was performed on the training set and used to transform the test set.

Modeling

Three models were trained and tested in the modeling step. Each was selected for specific reasons and diversity in how the models process the data. This allowed me to evaluate if increasing modeling complexity improved predictive power.

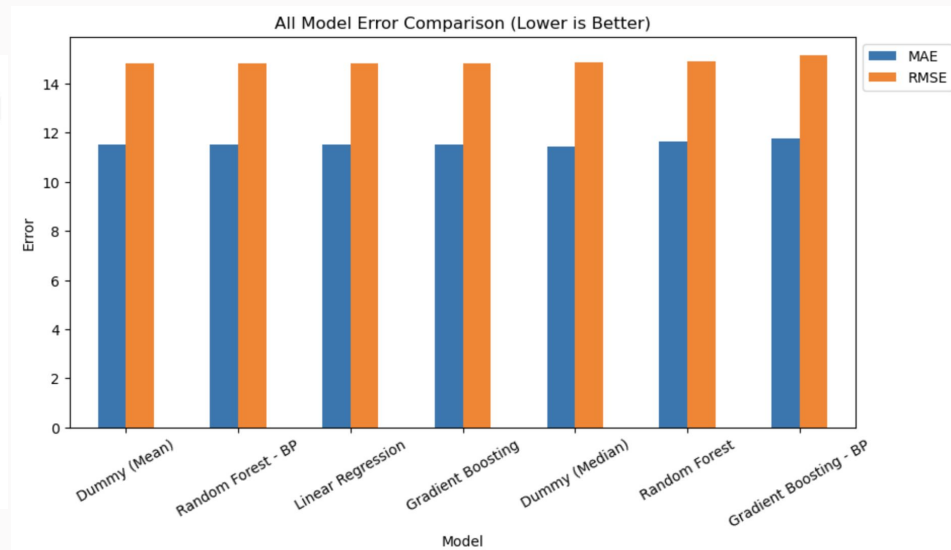
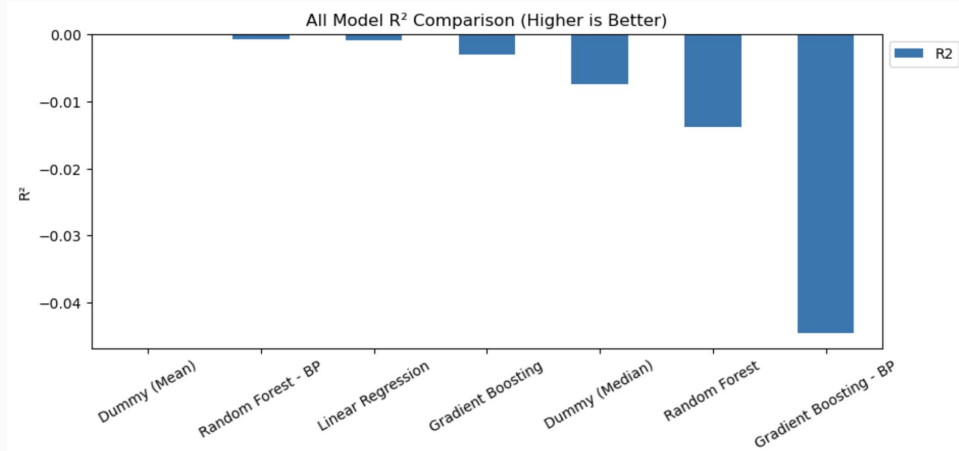
Linear Regression - Selected as a foundation model to detect if there are strong linear relationships present.

Random Forest Regression - Selected to capture non-linear relationships and interactions

Gradient Boosting Regression - Selected to test a high-capacity ensemble model

Two DummyRegressor models (mean and median strategies) were added to establish a baseline. These models predict the average popularity for all songs without learning from features.

Modeling



Trained models scored a negative R Squared score indicating the models could not explain variance in popularity based on the available features.

Alternative Approaches

Create Additional Signal

Artist Song Count was added to test if additional signal would produce a different result.

There was no significant change

Pivot to predict “Hit”

A Classification column was added to classify a son as a hit or not a hit based on a threshold set in popularity score. Several thresholds were tested.

Models showed accuracy paradox issues consistently guessing the same value to achieve highest accuracy rating.

Pivot to predict Country

Pivoted to predict the country the song will be popular in vs overall popularity. All supplemental country data had to be dropped to avoid leakage.

Same results as original model.

Conclusion

The conclusion from these tests is clear:

The current dataset does not contain enough pre-release information to accurately predict a song's success.

Additional Data that could be useful:

- Significantly larger dataset containing more songs per artist and a wider range of popularity
- More artist level features such as artist popularity, number of albums and songs released, home country, country of residence, other sources of fame
 - Additional song level features such as language of lyrics, band type/instruments present on track, recording type (professional studio vs home recorded etc)
- Additional Spotify Analytics Data such as number of saves, number of playlists, number of skips, etc.